

1. We can know that $4^3 = 64$ and $5^3 = 125$, since $64 < 82 < 125$. We know that the result is 4.
2. By the definition, the integer part of $\sqrt[r]{n}$ will be root, so that the root will equal to $\lfloor \sqrt[r]{n} \rfloor$.
3. Since $n > 0$, it isn't significant for us to guess $g < 0$. Or g^r may be negative. In addition, we also don't need to guess $g > n$. Because if $g > n$, g^r must be bigger than n . It is useless.
4. LowEnough should be 0 and tooHigh should be $n+1$.
5. The question will be is the guess number 5 is smaller than 47226. LowEnough should be 0 and tooHigh should be 47227. And then we can use $(0 + 47227)/2$ to guess and until $\text{lowEnough} + 1 = \text{tooHigh}$. But in my opinion, the 20^5 is 320000. If we use 0 and 20 will be better.
6.

```
private static int root(int n, int r) {
    int lowEnough = 0;
    int tooHigh = n+1;
    while ( lowEnough + 1 < tooHigh ){
        int guess = (lowEnough + tooHigh) / 2;
        if (power(guess, r) <=n){
            lowerEnough = guess;
        }else{
            tooHigh = guess;
        }
    }
}
```

```
}
```

```
return lowEnough;
```

```
}
```